

ORIGINAL ARTICLE

Application and effectiveness of adaptive AI in elderly healthcare

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Abstract

Background: In addressing elderly healthcare issues, cognitive impairment can cause significant disruptions in daily life and may potentially develop into dementia. Thus, finding ways to delay the progression of cognitive impairment is a critical issue.

Methods: This study aims to develop an adaptive artificial intelligence (AI) mechanism that creates enjoyable and beneficial content to help delay cognitive impairment in the elderly. Utilising virtual reality (VR) and a fishing game, the design enhances reaction time and attention through interactive fishing activities. The AI personalises content based on individual performance to improve cognitive function.

Results: Experimental results showed that adaptive AI increased participant satisfaction from 86.84 to 91.05 points and future willingness from 75.26 to 85.68 points. The number of fish caught rose from 98 to 120, with the average per participant increasing from 2.64 to 2.85.

Conclusions: This is undoubtedly the trend of the future. VR allows the elderly to have a more impactful and memorable first experience, while AI dynamically adjusts the game's difficulty based on the elderly's performance, addressing the issue of reduced willingness to continue due to inappropriate game difficulty. The VR game developed in this study is designed to be relaxing and incorporates mechanisms to promote the elderly's health. It is not restricted by location or time and, more importantly, meets the health promotion needs of the elderly.

Key words: adaptive AI, elder healthcare, virtual reality.

INTRODUCTION

With advances in medical technology and public health, demographic structures are gradually changing. The accelerated ageing of the global population has resulted in a marked increase in age-related diseases, with dementia identified as one of the most prevalent and concerning health challenges.¹ An estimated 55 million people worldwide are currently living with dementia, according to a report by the World Health Organization.¹ Approximately 10 million people are newly diagnosed with dementia annually.^{1,2} This figure is expected to surge to 78 million by 2030.³ Dementia leads to cognitive decline, affecting memory, judgement, and spatial orientation, often accompanied by delusions and hallucinations.^{3–6} The

high cost of dementia care places a significant financial burden on patients and their families.¹ Although there is currently no cure for dementia, both the American Neurological Association and the UK's Alzheimer's Research Foundation suggest that dementia can be prevented.^{2,7–9} Promoting brain activity and reducing the risk of dementia have emerged as critical research priorities.⁴

A study featured in *The Economist* revealed that engaging in crossword puzzles can slow memory decline by 2.54 years.¹⁰ An Australian 20-year study found that playing computer games, such as Tetris, helps retirees at risk of dementia maintain cognitive agility. The American Neurological Association also noted that mental inactivity can lead to dementia in

the elderly, and activities like playing puzzle games and solving riddles can help reduce its incidence.^{3,11–13} Studies in epidemiology, gerontology, and psychology have consistently demonstrated a positive link between regular engagement in board or computer games and the health of elderly individuals.^{14–19} As a result, numerous organisations and research institutions are encouraging the elderly to engage in tabletop and mobile games to help prevent the onset of dementia. However, traditional methods often struggle to attract and sustain the interest of elderly individuals, necessitating more dynamic and engaging approaches.

Virtual reality (VR) technology, known for creating immersive experiences, is valuable in fields like medical training, education, and rehabilitation.^{20–23} This study utilises VR due to its ability to surpass traditional devices by providing more engaging and stimulating environments. The higher levels of immersion offered by VR create experiences that feel more meaningful and enjoyable, which is particularly crucial for elderly individuals, especially those who are homebound. Unlike traditional, often monotonous activities, VR offers interactive and dynamic content that feels less like therapy and more like an enjoyable experience. The goal is to promote health through engaging and stimulating activities.

Given the diverse conditions of elderly individuals, it is challenging to ensure equal enjoyment of the VR experience. This is where artificial intelligence (AI) plays a crucial role in adapting the VR environment to meet each user's needs. AI, commonly used for big data analysis and predictive modelling, plays a key role in assessing and adjusting game difficulty based on user performance.^{24–26} By continuously monitoring interactions, AI personalises the experience, ensuring it remains enjoyable and appropriately challenging for each user. Instead of focusing on isolated actions, AI analyzes ongoing gameplay to identify patterns and maintain a balanced challenge, improving the overall experience for elderly users.^{27,28}

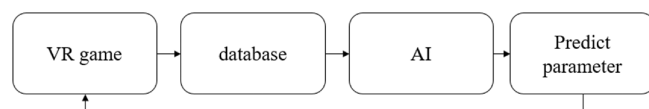


Figure 1 The overall architecture of an artificial intelligence (AI)-adapted virtual reality (VR) environment.

This study integrates AI and VR technologies to combat dementia by creating interactive, engaging content for the elderly. Using advanced VR equipment, we provide stimulating experiences that encourage participation. AI dynamically adjusts the difficulty based on individual conditions, ensuring enjoyable gameplay. This approach promotes sustained engagement and contributes to the health and well-being of the elderly.

METHODS

Design of an AI-adapted VR environment

This study developed an AI-adapted VR environment specifically designed for the elderly. The game aims to provide different experiences and stimulation while promoting physical health. To quickly engage the elderly in the game content, the study considers various factors such as individual physical condition, emotional state, and immediate reactions. By leveraging AI to adjust the game content and difficulty, we ensure that the game is tailored to each elderly individual. This personalised approach not only facilitates quicker immersion and understanding of the game but also encourages elderly to continue long-term training to improve their responsiveness.

Design architecture and model

By utilising AI to analyze the relationship between the elderly performance and game parameters, the game adjusts these parameters based on individual performance. This approach is similar to adaptive learning,^{29–32} where the system caters to the varying learning speeds and abilities of different elderly persons. However, it provides personalised teaching methods through computational support, ensuring that all learners enjoy the process and achieve optimal results.

The design process has four parts, as shown in Figure 1. The elderly interact with the VR game using VR equipment, generating gameplay states that are stored in a database along with current game parameters. During training, an AI model retrieves and pre-processes this data for analysis and training. Once trained, the model predicts game parameters based on new elderly states, which the VR game integrates for further gameplay. This iterative process adjusts the difficulty to match the elderly person's needs.

This paper focuses on parameter training and adjustment through a multi-dimensional input and output approach, using the user's in-game state as input parameters and their actions and reactions as output parameters. The input parameters are divided into four stages:

- 1 Initial stage: age.
- 2 Bait-biting stage: consists of handle displacement, user reaction time, and success status.
- 3 Line-reeling stage: encompasses maximum time in the green zone, average meter value, times exceeding and below the green zone, duration, reel status, and success in catching fish.
- 4 Directional interaction stage: involves handle displacement, user reaction time, and success status.

The output parameters are processed based on the input parameters at each stage. Potential parameters in the bait-biting stage include the minimum handle displacement required and maximum user reaction time. In the line-reeling stage, they include meter increase speed, meter decay speed, thresholds for exceeding and being below the green zone, fish stamina reduction, fish struggle duration, and overall fish strength. In the directional interaction stage, they include the minimum handle displacement required and maximum user reaction time, totalling 11 interaction parameters.

Adaptive AI algorithm

This study designed a content incorporating reaction and attention tests under the concept of adaptability. The adaptive application ensures that the test

content adjusts appropriately to the elderly. After adjustment, the test content avoids causing excessive training fatigue, anxiety from overly high difficulty, or boredom from too low difficulty. The content is personalised to provide an enjoyable and effective training experience for the elderly.

Elderly performance is broadly evaluated using spatial relationships, temporal relationships, and overall performance. Spatial tasks require specific movements, with difficulty assessed by the discrepancy between the elderly and target movements. Temporal tasks are time-sensitive, with difficulty increasing as the time limit shortens. A combination of spatial and temporal tasks may require specific movements within a time limit. Task success rate provides a numerical representation of the elderly performance, summarising past experiences across different difficulty levels.

The sequential evaluations include success rate, interaction time, and interaction displacement, as illustrated in Figure 2 and Algorithm 1. In the fishing interaction mechanism, the goal is to deplete the fish's stamina in order to catch it. The first criterion checks if the elderly interact with the same fish two times or less and if the interaction duration is short, indicating low difficulty. The second criterion checks if the elderly interact with the fish more than twice but less than four times, with a success rate higher than 0.25, indicating successful catching within four attempts.

The criteria evaluate conditions for catching fish. If the fish is not caught, we assess the elderly success

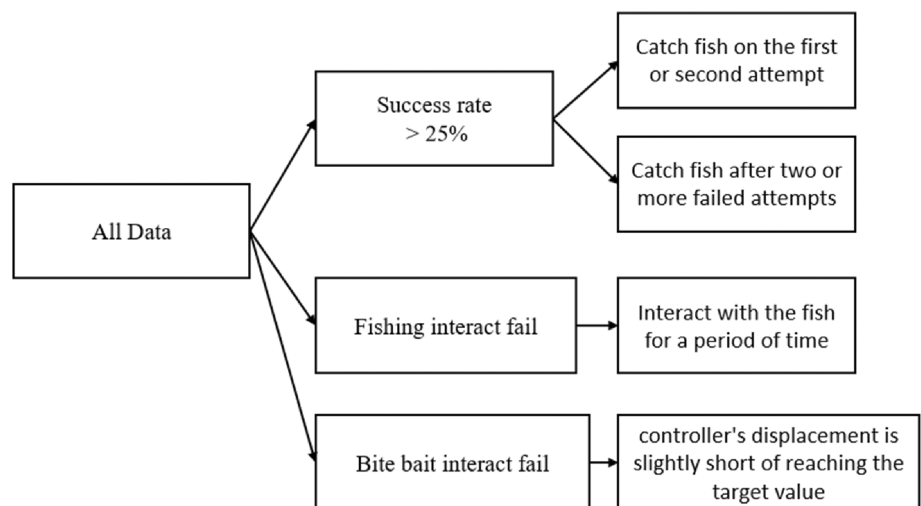


Figure 2 The process of filtering data.

Algorithm 1

AI-ADAPTED ALGORITHM

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Input    original data
Output   after filtering data
1 Success Rate = Input[Get] / (Input[Get] + Input[Fail])
2 IF(Success Rate > = 0.25) then
3   IF((Input[Fail] < 2) and (Input[Fish Up Time]
   > = 20)) do save Output
4   IF(Input[Fail] > = 2) do save Output
5 IF(Input[Is Get] == 0 and Input[Up Time] > = 20)
  do save Output
6 IF((Input[Is Triggered] == 0) and (Input[elderly
  Distance] / Input[need Distance] > = 0.8))
7 do save Output

```

proximity based on interaction time and action displacement. Long reeling times indicate a close attempt hindered by timeout or line break, while insufficient hand movement during hooking suggests a near miss. Adaptive rules aim to match the elderly state variables with appropriate game variables, focusing on potential success rather than difficulty extremes. Filtered data are then standardised for AI training.

AI model evaluation

(1) *Pilot test experiment.* The research team recruited participants for a pilot test to serve as a control group and collect data for AI training. Each participant's experiment lasts 10 min: 15 min for tutorial

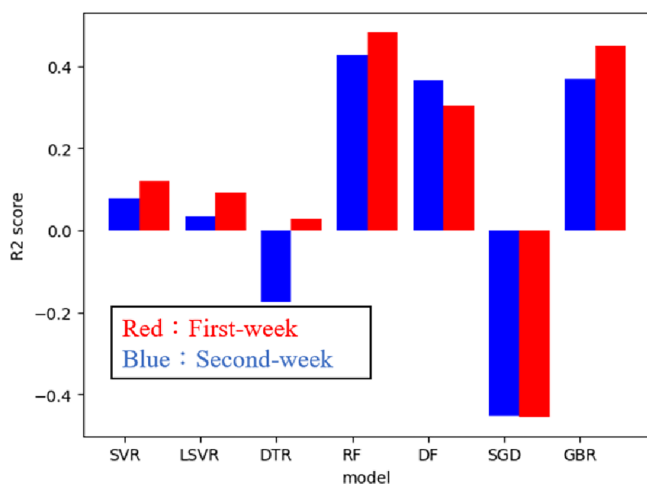


Figure 3 R2Score status of each model in 2 weeks.

gameplay and 5 min for fishing operations. Participants complete a feedback questionnaire afterward. The difficulty of the interactive content remains unchanged, with game parameters randomly generated and the tutorial is identical to actual gameplay. Data collection starts during the tutorial, resulting in 638 data points. These are split into 510 for the training set and 128 for the test set. The AI model's performance is compared as the data amount increases.

This research selects the AI model for the finetune test by comparing R2Score and MAE (Mean Absolute Error). MAE, also known as L1 loss or L1 norm loss, is used to evaluate the performance of AI models. It calculates the error between the true values and predicted values, takes the absolute value of these errors, and sums them up. This approach avoids the cancellation effect of positive and negative values, ensuring the error measurement reflects the true situation. Therefore, the MAE provides a better representation of the prediction error performance. The AI models tested include SVR (Support Vector Regression),³³ LSVR (Linear Support Vector Regression),³⁴ DTR (Decision Tree Regression),^{35,36} RF (Random Forest),^{37,38} DF (Deep Forest),³⁹ SGD (Stochastic Gradient Descent),^{40,41} and GBR (Gradient Boosting Regressor).⁴² Figures 3 and 4, along with Table 1, show RF achieved the highest R2Scores consistently over 2 weeks. Although DF outperformed RF in MAE during the first week, RF surpassed DF in the second week. RF's

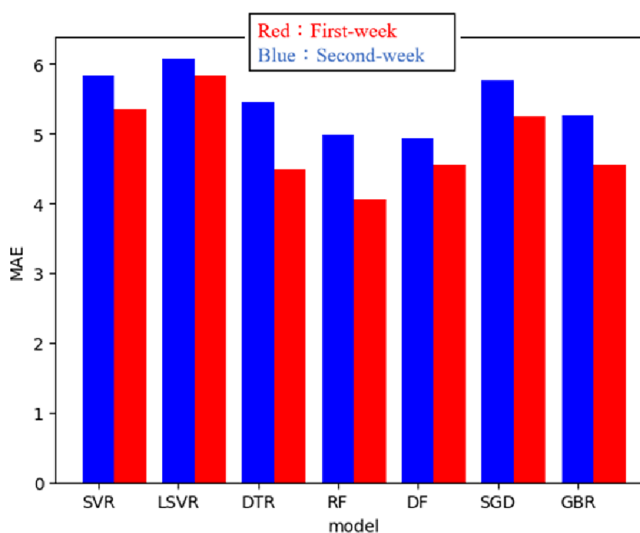


Figure 4 Mean Absolute Error status of each model in 2 weeks.

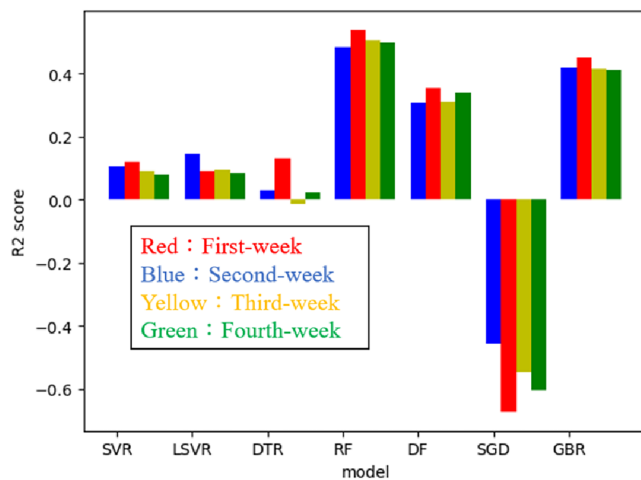
Table 1 Comparison of actual values of each model in 2 weeks

Model	First week		Second week		Comparison	
	R2Score	MAE	R2Score	MAE	R2Improve (%)	MAE (%)
SVR	0.0782	5.83	0.1192	5.35	52.429 67	−8.233 28
LSVR	0.0323	6.08	0.092	5.84	184.8297	−3.947 37
DTR	−0.1859	5.45	0.0289	4.48	−115.546	−17.7982
RF	0.4259	4.99	0.483	4.06	13.4069	−18.6373
DF	0.3666	4.94	0.3051	4.55	−16.7758	−7.894 74
SGD	−0.4521	5.76	−0.4551	5.24	0.663 57	−9.02778
GBR	0.3686	5.27	0.4497	4.55	22.00217	−13.6622

Abbreviations: DF, deep forest; DTR, decision tree regression; GBR, gradient boosting regressor; LSVR, linear support vector regression; MAE, mean absolute error; RF, random forest; SGD, stochastic gradient descent; SVR, support vector regression.

Table 2 Amount of data per week

	First week	Second week	Third week	Fourth week
Data	638	829	907	990

**Figure 5** R2Score status around each model.

improvement rate is lower than other models but already reaches a high standard, making RF the chosen model for the finetune test.

(2) *Finetune test experiment.* The finetune test examines AI's impact on user experience. AI adjusts game variables based on the player's state, with difficulty changes during game loading or when catching a fish. Each experiment lasts 10 min: 15 min for instruction and 5 min for gameplay, followed by a questionnaire. The process mirrors the pilot test but incorporates AI. Spanning 3 weeks, each week's model trains on data from the previous week. This multi-week approach allows for better observation of model changes. Weekly data totals are shown in

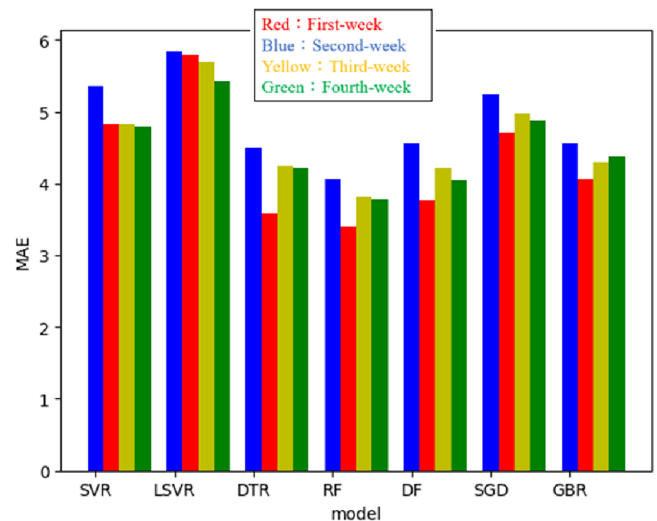
**Figure 6** Mean Absolute Error status around each model.

Table 2, represented by different colours for each week.

DTR showed significant improvement in R2Score and a notable decrease in MAE, as seen in Figures 5 and 6, Table 3. Although some models were expected to surpass RF, Table 4 shows dataset fluctuations decreased model generalisation and accuracy, possibly due to new participant characteristics or insufficient dataset size. Table 5 indicates RF's R2Score slightly declined and MAE slightly increased, suggesting model stabilisation.

Cross-validation was conducted by dividing the dataset into five parts, each serving as a training and test set in turn. The average R2Score was 0.465, with a maximum of 0.503 and a minimum of 0.409. Compared to the final RF performance on the test set (R2Score of 0.4978 in Table 6), the average score differed by 0.0328. This suggests some variation in

Table 3 Comparison of actual numerical values of each model (week: first and second)

Model	First week		Second week		Comparison	
	R2Score	MAE	R2Score	MAE	R2Improve (%)	MAE (%)
SVR	0.1062	5.35	0.1192	4.83	12.24	−9.69
LSVR	0.1454	5.84	0.0920	5.79	−36.74	−0.84
DTR	0.0289	4.48	0.1296	3.57	348.24	−20.36
RF	0.4830	4.06	0.5375	3.39	11.29	−16.57
DF	0.3051	4.55	0.3535	3.75	15.86	−17.52
SGD	−0.4551	5.24	−0.6714	4.70	−47.52	−10.33
GBR	0.4200	4.55	0.4497	4.06	7.09	−10.69

Abbreviations: DF, deep forest; DTR, decision tree regression; GBR, gradient boosting regressor; LSVR, linear support vector regression; MAE, mean absolute error; RF, random forest; SGD, stochastic gradient descent; SVR, support vector regression.

Table 4 Comparison of actual numerical values of each model (week: second and third)

Model	Second week		Third week		Comparison	
	R2Score	MAE	R2Score	MAE	R2Improve (%)	MAE (%)
SVR	0.1192	4.83	0.0906	4.82	−23.98	−0.16
LSVR	0.0920	5.79	0.0956	5.69	3.95	−1.77
DTR	0.1296	3.57	−0.0119	4.24	−109.21	18.74
RF	0.5375	3.39	0.5042	3.82	−6.19	12.6
DF	0.3535	3.75	0.3103	4.21	−12.23	12.05
SGD	−0.6714	4.70	−0.5466	4.97	18.59	5.69
GBR	0.4497	4.06	0.4165	4.30	−7.39	5.83

Abbreviations: DF, deep forest; DTR, decision tree regression; GBR, gradient boosting regressor; LSVR, linear support vector regression; MAE, mean absolute error; RF, random forest; SGD, stochastic gradient descent; SVR, support vector regression.

Table 5 Comparison of actual numerical values of each model (week: third and fourth)

Model	Third week		Fourth week		Comparison	
	R2Score	MAE	R2Score	MAE	R2Improve (%)	MAE (%)
SVR	0.0906	4.82	0.0792	4.79	−12.54	−0.71
LSVR	0.0956	5.69	0.0825	5.42	−13.78	−4.76
DTR	−0.0119	4.24	0.0243	4.20	303.79	−0.9
RF	0.5042	3.82	0.4978	3.77	−1.28	−1.17
DF	0.3103	4.21	0.3401	4.04	9.61	−3.99
SGD	−0.5466	4.97	−0.6041	4.87	−10.53	−1.96
GBR	0.4165	4.30	0.4096	4.37	−1.66	1.73

Abbreviations: DF, deep forest; DTR, decision tree regression; GBR, gradient boosting regressor; LSVR, linear support vector regression; MAE, mean absolute error; RF, random forest; SGD, stochastic gradient descent; SVR, support vector regression.

Table 6 Cross-validation R2Score of each model

Model	First	Second	Third	Fourth	Fifth	Average
SVR	0.102	0.068	0.073	0.091	0.07	0.081
LSVR	0.132	0.087	0.092	0.121	0.095	0.105
DTR	0.007	0.026	−0.095	0.047	−0.073	−0.018
RF	0.499	0.503	0.409	0.48	0.433	0.465
DF	0.326	0.349	0.307	0.337	0.311	0.326
SGD	−0.482	−0.354	−0.582	−0.361	−0.446	−0.445
GBR	0.415	0.424	0.351	0.403	0.358	0.39

Abbreviations: DF, deep forest; DTR, decision tree regression; GBR, gradient boosting regressor; LSVR, linear support vector regression; MAE, mean absolute error; RF, random forest; SGD, stochastic gradient descent; SVR, support vector regression.

model performance across different datasets, indicating a need for further dataset classification.

EXPERIMENTS

Participants

In this study, participants aged 60 and above were randomly recruited from various locations. The pre-test included 38 participants (21 males, 17 females) with an average age of 67.86 and a standard deviation (SD) of 6.45. The post-test included 51 participants (21 males, 30 females) with an average age of 67.36 and an SD of 6.90. Nine participants (five males, four females) took part in both tests, averaging 65.22 years with an SD of 5.38. Overall, 89 individuals participated in either the pre-test or post-test, with nine participating in both.

Data collection and analysis

This experiment included pre-test and post-test phases. In the pre-test, participants engaged with VR content without AI-driven dynamic difficulty adjustment, with difficulty levels randomly generated to collect data for AI training. After training the model, the post-test phase involved participants playing the VR content with AI-enabled dynamic difficulty adjustment. The process is illustrated in Figure 7.

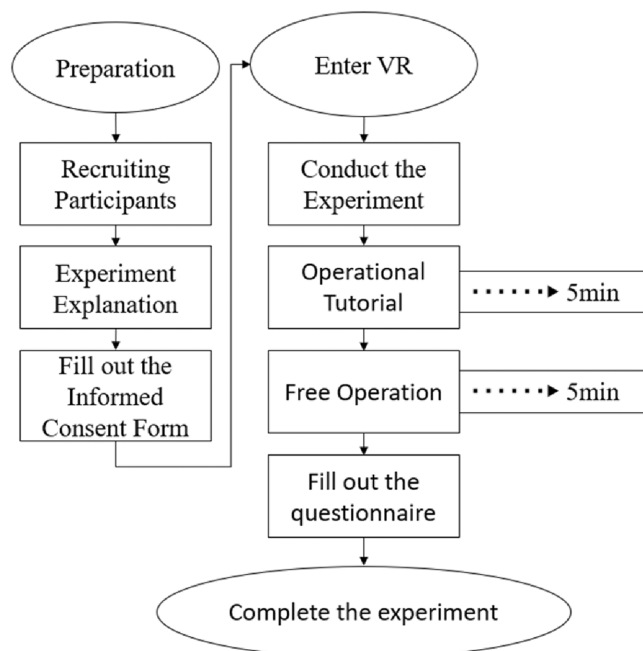


Figure 7 Experimental procedure.

Instruments and activities

The experimental procedure is illustrated in Figure 7. Below is an explanation of each step, detailing how the process is conducted, what activities take place, and how the experiment is ultimately completed.

- 1 Finding participants: participants for this study were not recruited through any platforms. Instead, researchers personally visited areas where individuals aged 60 and above are likely to be found, such as parks, schools, and so forth, to seek volunteers willing to participate in the experiment.
- 2 Experiment explanation: the researchers provided the participants with detailed instructions on the upcoming procedures and what to expect.
- 3 Signing the informed consent form: after understanding the experimental procedures and confirming their physical and mental well-being, participants will sign the informed consent form, allowing them to begin the experiment.
- 4 Conducting the experiment: after assisting participants with the VR equipment, the researchers will proceed with a 10-min experiment.
- 5 Instruction session: the researchers will assist participants with a 5-min interactive tutorial, during which data will be collected for AI training purposes.
- 6 Free exploration: after completing the instructional session, participants will begin freely exploring and attempting to complete the VR interactive tasks within a 5-min period.
- 7 Completing the questionnaire: after finishing the 10-min interactive experiment, participants will complete a questionnaire, marking the conclusion of the entire experimental process.

Questionnaires

The questionnaire is divided into seven sections based on the Delone and McLean IS success model.⁴³ The first section collects personal information such as experiment ID, age, gender, education level, and gaming experience. The remaining six sections consist of multiple-choice questions with options from 'Strongly Agree' to 'Strongly Disagree,' scored from 5 to 0 for statistical analysis. These sections cover software and hardware responsiveness, comfort of software content, whether the software meets expectations, contribution to health promotion, user satisfaction, and intention to use the software again.

RESULTS AND DISCUSSION

This study uses AI to adjust the difficulty of interactive content based on the elderly performance, aiming to provide an enjoyable experience and promote health through gameplay. We will analyze how AI adjusts difficulty and understand the elderly's feedback via questionnaires. In the pre-test, difficulty was random. In the post-test, AI-generated game variables are based on past gameplay for returning participants. For first-time players, the initial difficulty is random until a fish is caught or the game restarts, after which AI adjusts difficulty based on their performance.

Elderly performance analysis.

Table 7 shows participants' performance in both pre-test and post-test, with average interaction duration measured from when a fish bites the bait. The study found that post-test participants caught more fish and had longer interaction durations due to AI-adjusted difficulty. Each time a fish was caught, the game recalibrated to match the elderly performance, increasing difficulty and interaction duration as more fish were caught.

These bases cannot accurately reflect the rationality of difficulty changes. This study uses reaction time

and attention as health indicators, aiming to promote health by training these faculties. The VR content includes tests requiring significant reaction time and attention, making perceived difficulty highly correlated with these factors. Factors like different testing locations or elderly fatigue can influence performance. Since the pre-test and post-test are sequential, they are not directly comparable. Improved performance in the post-test may also result from increased familiarity with the equipment and better understanding from the pre-test.

In summary, this study investigates whether the difficulty has become easier or harder. First-time players in both pre-test and post-test were compared. Table 8 shows that the number of fish caught increased from 98 to 120, with the average per participant rising from 2.64 to 2.85. Including re-testers, the total increased to 153, averaging three per participant. Despite a higher age SD in the post-test (7.01 vs. 6.40), the results showed an increase in fish caught, suggesting the level of difficulty has become easier, making it easier for participants to catch fish.

In discussing increased difficulty, it is necessary to measure it under the same test environment and time point. This study examines the impact of AI-

Table 7 Comparison of performance of repeaters on adaptive methods in pre- and post-tests

Participants	Age	Pre-test		Post-test	
		Average interaction time (seconds)	Fish caught (number)	Average interaction time (seconds)	Fish caught (number)
T2	76	17.81	1	20.75	2
T3	67	20.12	1	45.07	4
T5	63	24.99	1	54.84	5
T6	60	18.65	1	29.64	5
T17	71	18.80	0	23.95	3
T18	62	17.89	0	28.13	0
T22	66	37.37	1	35.56	1
T32	60	16.17	1	20.91	2
T33	60	47.20	3	54.73	4

Table 8 Comparison of the number of fish caught pre-test and post-test

	Pre-test	Post-test	Post-test (contain repeat)
Total	37	42	51
Male	20	17	21
Female	17	25	30
Age average	68.08	69.5	67.36
Age standard deviation	6.40	7.01	6.90
Total number of fish caught	98	120	153
Average catch per person	2.64	2.85	3

integrated difficulty adjustments in the post-test. Difficulty increases at the start of interaction and when a fish is caught, it is considered a good performance. This phenomenon is observed when participants catch many fish. The study investigates if the time taken to catch a fish increases, indicating rising difficulty for participants who catch three or more fish in the post-test. Detailed results in Table 9 show

Table 9 Post-test to observe the fishing time and learn about changes in difficulty

Participants	The time it takes to catch a fish (seconds)				
	First	Second	Third	Fourth	Fifth
T3	22.74	19.70	41.23		
T5	38.25	45.17	55.39	76.55	79.19
T6	44.22	28.27	24.23	33.63	13.18
T17	35.90	46.73	28.13		
T49	17.70	40.17	53.52		
T51	69.99	21.36	14.20	21.45	
T53	100.13	73.88	48.43		
T55	17.00	35.17	49.37		
T56	23.48	19.15	40.45		
T59	46.49	15.90	35.22		
T60	21.43	76.81	32.13		
T61	13.97	19.90	31.57		
T70	38.03	62.97	69.89		
T71	26.05	54.79	39.44		
T73	65.70	5.51	32.90		
T74	56.81	5.34	9.29		
T77	26.09	2.92	45.08	47.5728	
T78	23.88	17.41	54.80	52.61	
T83	54.56	16.64	28.64	30.17	
T86	16.39	22.65	67.13		

Table 10 Participant information

	Males (pre-test + post-test)	Females (pre-test + post-test)
Total	43	46
Standard deviation age	5.62	7.52
Average age	67.48	69.19

Table 11 Pre-test and post-test for males, and pre-test and post-test for females

Units	Males (pre-test + post-test)	Females (pre-test + post-test)	Difference (female - male)
System quality	87.25	83.39	-3.86
Information quality	93.02	93.76	+0.74
Expected value	88.37	87.36	-0.98
Health function	94.72	92.60	-2.12
Satisfaction	87.90	91.01	+3.10
Future use intention	82.17	79.13	-3.04
Average	88.74	87.43	-1.30

20 participants caught three or more fish, including four repeat testers and 16 first-time players.

Among the four repeat testers, only T5 showed a clear increasing trend. Among the 16 first-time participants, most displayed an increasing trend after the second catch, including T49, T55, T56, T59, T61, T70, T73, T74, T77, T78, T83, and T86. Since the initial difficulty for first-time players is randomly generated, the first interaction does not reflect AI-adjusted difficulty. At the first time, the elderly showed more noticeable difficulty with adjustments compared to repeat testers, possibly because AI adjustments for repeat testers referenced pre-test data where participants were unfamiliar with the gameplay or fatigued, affecting their performance and leading to inaccurate AI adjustments.

User satisfaction analysis

After experiencing the VR interactive content, participants completed a questionnaire categorised into six types: system quality, information quality, expectations, health functionality, satisfaction, and future intention. Each item is rated from 1 to 5 points. The total score for each category is converted into a percentage based on the maximum possible score for that category. For example, a score of 15 out of a maximum of 25 points for system quality would be $(15/25) * 100 = 60\%$.

Tables 10 and 11 compare the perspectives of males and females on interactive content in pre-test and post-test assessments. Males show a larger score gap in system quality and equipment

Table 12 Male participants information

	Pre-test	Post-test
Total	21	22
Standard deviation age	6.05	5.28
Average age	68	67

Table 13 Comparison of males in pre-test and post-test

Units	Pre-test	Post-test	Difference (post-test – pre-test)
System quality	87.42	87.09	–0.33
Information quality	93.33	92.72	–0.60
Expected value	86.66	90	+3.33
Health function	96.19	93.33	–2.85
Satisfaction	84.44	91.21	+6.76
Future use intention	74.60	89.39	+14.79
Average	87.14	90.27	+3.12

Table 14 Female participants information.

	Pre-test	Post-test
Total	17	29
Standard deviation age	7.09	7.75
Average age	67.70	70.06

Table 15 Comparison of females in pre-test and post-test

Units	Pre-test	Post-test	Difference (post-test – pre-test)
System quality	85.64	82.06	–3.57
Information quality	95.68	92.64	–3.04
Expected value	90.19	85.74	–4.44
Health function	97.25	89.88	–7.36
Satisfaction	89.80	91.72	+1.92
Future use intention	76.07	80.91	+4.84
Average	88.76	86.65	–2.10

Table 16 Comparison of participants in pre-test and post-test

Units	Pre-test	Post-test	Difference (post-test – pre-test)
System quality	86.63	86.41	–3.58
Information quality	94.38	92.93	–1.84
Expected value	88.24	88.67	–1.26
Health function	96.66	92.43	–5.87
Satisfaction	86.84	91.05	+4.90
Future use intention	75.26	85.68	+8.70
Average	88.10	89.22	–0.44

operation, likely due to greater familiarity with VR controllers from gaming. This gap may also be influenced by a higher SD among females, as older individuals have fewer gaming opportunities. However, there is no significant gender difference in information quality or expectations for the game.

Health functionality scores higher for males, likely due to the novelty of health-promoting games. Satisfaction is higher for females, possibly because they play games less frequently and have fewer comparisons. Consequently, males show a higher future intention to engage with the game.

Tables 12 and 13 show that males had a better post-test experience. In contrast, Tables 14 and 15 indicate females experienced more negative growth, possibly due to higher expectations for AI integration. Males exhibited noticeable negative growth in health functionality, likely due to the younger and more educated post-test participants, who prefer outdoor activities over health-promoting equipment. This does not suggest they would avoid such equipment as they age.

Table 16 shows that the post-test with AI integration performed worse in several categories compared

Table 17 Comparison of repeat participants in pre-test and post-test

Units	Pre-test	Post-test	Difference (post-test – pre-test)
System quality	93.33	89.77	–3.55
Information quality	99.25	93.33	–5.92
Expected value	94.81	90.37	–4.44
Health function	100	94.07	–5.92
Satisfaction	88.14	90.37	+2.22
Future use intention	74.07	87.40	+13.33
Average	91.77	90.77	–1

to the pre-test, likely due to higher expectations for AI as indicated in Table 15. Interestingly, Table 17, focusing on repeat testers, supports this finding. Despite this, Table 13 (males), 15 (females), 16 (all participants), and 17 (repeat testers) reveal higher satisfaction and a greater willingness to continue using the equipment for interactive content in the post-test with AI integration.

CONCLUSION

With an increasing average lifespan, an ageing society is inevitable, bringing more attention to elderly health. This study focuses on mitigating dementia risks, which literature suggests can be delayed through activity, thinking, and new stimuli. Given that most the elderly activities are static, we chose fishing as the basis for a game designed to train reaction time and attention, engaging the hippocampus and aiding memory. This game provides a novel option for promoting elderly health.

Additionally, we have integrated ageing issues with emerging VR and AI technologies, which represent the future trend. VR offers a memorable first experience, while AI adjusts game difficulty based on the elderly performance, enhancing engagement. The VR game developed in this study is designed to be relaxing and promotes health, unrestricted by location or time, and meets the health needs of the elderly.

Participants were exclusively elderly individuals recruited randomly to better reflect general views and enhance the experiment's credibility. Questionnaire results showed participants rated the software's health functions above 90 points. AI integration made it easier for beginners, increasing satisfaction from 86.84 to 91.05 and future usage willingness from 75.26 to 85.68. This study not only promotes health but also aims to engage the elderly through

unique VR gaming experiences and adjustable AI difficulty, enhancing sustained health promotion.

While these findings are promising, there are opportunities for further improvement in real-world applications. Expanding the sample size and including more diverse participants will enhance the generalisability of the results. Finally, refining AI algorithms to support a wider range of cognitive and physical abilities will ensure greater adaptability and effectiveness for diverse users.

FUTURE WORK

This study did not specifically distinguish participants based on medication use, such as antidepressants. Future research could explore the effects of adaptive AI and VR interventions on elderly individuals who are on medication by comparing cognitive performance and improvements between medicated and non-medicated groups.

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DISCLOSURE

The authors have no potential conflicts of interest to disclose.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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